

CBSE Class 12 Maths — Differential Equations

50 Most-Probable PYQs & Board-Pattern Practice Questions

1-Mark (MCQ/Assertion-Reason/VSA)

- Q1.** Determine the order and degree (if defined) of the differential equation: $\left(\frac{d^2y}{dx^2}\right)^2 + \cos\left(\frac{dy}{dx}\right) = 0$. [PYQ 2020] [Confidence: Very High]
- Q2.** Find the integrating factor of the linear differential equation: $\frac{dy}{dx} + \frac{y}{x} = x^2$. [PYQ 2019] [Confidence: High]
- Q3.** Write the degree of the differential equation: $x\left(\frac{d^2y}{dx^2}\right)^3 + y\left(\frac{dy}{dx}\right)^4 + x^3 = 0$. [PYQ 2018] [Confidence: Very High]
- Q4.** What is the general solution of the differential equation $\frac{dy}{dx} = \frac{y}{x}$? [PYQ 2017] [Confidence: High]
- Q5.** Write the integrating factor (I.F.) for the linear differential equation: $x\frac{dy}{dx} - y = 2x^3$. [PYQ 2022] [Confidence: Very High]
- Q6.** State the order of the differential equation that represents the family of all concentric circles centered at the origin. [Practice] [Confidence: Medium]
- Q7.** Determine the degree of the differential equation: $\left(\frac{dy}{dx}\right)^3 + \frac{d^2y}{dx^2} + \sin\left(\frac{dy}{dx}\right) = 0$. [PYQ 2023] [Confidence: Very High]
- Q8.** Write the integrating factor for the differential equation: $\frac{dy}{dx} + y \sec x = \tan x$. [PYQ 2019] [Confidence: High]
- Q9.** Write the order and degree of the differential equation: $\frac{d^2y}{dx^2} = \sqrt{1 + \left(\frac{dy}{dx}\right)^2}$. [PYQ 2016] [Confidence: Very High]
- Q10.** What is the general solution of the differential equation $\frac{dy}{dx} = e^{x+y}$? [PYQ 2020] [Confidence: High]
- Q11.** Determine the order and degree (if defined) of the differential equation: $\left(\frac{d^3y}{dx^3}\right)^2 + \frac{dy}{dx} = e^x$. [PYQ 2018] [Confidence: Very High]
- Q12.** Find the integrating factor of the linear differential equation: $(1 - x^2)\frac{dy}{dx} - xy = 1$. [PYQ 2017] [Confidence: High]
- Q13.** Determine the order and degree of the differential equation: $\left(\frac{ds}{dt}\right)^4 + 3s\frac{d^2s}{dt^2} = 0$. [PYQ 2021] [Confidence: Very High]
- Q14.** Write the integrating factor for a linear differential equation of the form $\frac{dx}{dy} + Px = Q$, where P and Q are functions of y . [Practice] [Confidence: High]
- Q15.** Find the general solution of the differential equation: $\frac{dy}{dx} = \frac{x^2}{y^2}$. [PYQ 2023] [Confidence: Medium]
- Q16. Assertion (A):** The order of the differential equation formed by the family of curves $y = A \sin x + B \cos x$ is 2. **Reason (R):** The order of a differential equation representing a family of curves is equal to the number of arbitrary constants present in the equation of the curves. Choose the correct option. [Practice] [Confidence: Very High]
- Q17. Assertion (A):** The degree of the differential equation $y' + \sin(y'') = 0$ is 1. **Reason (R):** A differential equation must be a polynomial equation in derivatives for its degree to be defined. Choose the correct option. [PYQ 2024] [Confidence: High]
- Q18.** Find the integrating factor of the differential equation $x \log x \frac{dy}{dx} + y = \frac{2}{x} \log x$. [PYQ 2020] [Confidence: Very High]
- Q19.** State the order of the differential equation: $\frac{d^4y}{dx^4} + \sin\left(\frac{d^2y}{dx^2}\right) = 0$. [PYQ 2019] [Confidence: High]
- Q20.** What is the form of the substitution used to solve a homogeneous differential equation of the type $\frac{dx}{dy} = h\left(\frac{x}{y}\right)$? [Practice] [Confidence: High]

2-Mark

- Q21.** Solve the differential equation: $\frac{dy}{dx} = \frac{1+y^2}{1+x^2}$. [PYQ 2018] [Confidence: High]
- Q22.** Solve the differential equation: $\frac{dy}{dx} = e^{x-y} + x^2e^{-y}$. [PYQ 2019] [Confidence: Very High]
- Q23.** Solve the differential equation: $\frac{dy}{dx} = 1 + x + y + xy$. [PYQ 2020] [Confidence: High]
- Q24.** Find the general solution of the differential equation: $\frac{dy}{dx} = 2^{x+y}$. [PYQ 2017] [Confidence: Medium]
- Q25.** Find the integrating factor and the general solution of the linear differential equation: $\frac{dy}{dx} + y = e^{-x}$. [PYQ 2022] [Confidence: Very High]
- Q26.** Solve the differential equation by rearranging terms: $(x + y)\frac{dy}{dx} = 1$. [PYQ 2019] [Confidence: High]
- Q27.** Find the particular solution of the differential equation $\frac{dy}{dx} = -4xy^2$, given that $y = 1$ when $x = 0$. [PYQ 2020] [Confidence: Very High]
- Q28.** Show that the differential equation $x\frac{dy}{dx} = y(\log y - \log x + 1)$ is a homogeneous differential equation. [Practice] [Confidence: High]

Q29. Solve the differential equation: $\frac{dy}{dx} = \sqrt{4 - y^2}$, where $-2 < y < 2$. [PYQ 2016] [Confidence: Medium]

Q30. Find the general solution of the differential equation: $\sec^2 x \tan y dx + \sec^2 y \tan x dy = 0$. [PYQ 2018] [Confidence: Very High]

Q31. Solve the differential equation by substitution: $\frac{dy}{dx} = \sin(x + y)$. [PYQ 2021] [Confidence: High]

Q32. Form the differential equation representing the family of parabolas having their vertex at the origin and axis along the positive y-axis. [PYQ 2017] [Confidence: Medium]

3-Mark

Q33. Solve the linear differential equation: $(1 + x^2)\frac{dy}{dx} + 2xy = \frac{1}{1+x^2}$. [PYQ 2019] [Confidence: Very High]

Q34. Solve the homogeneous differential equation: $x\frac{dy}{dx} - y = \sqrt{x^2 + y^2}$. [PYQ 2020] [Confidence: Very High]

Q35. Solve the linear differential equation: $x\frac{dy}{dx} + 2y = x^2 \log x$. [PYQ 2022] [Confidence: High]

Q36. Find the particular solution of the differential equation $(x^2 - y^2)dx + 2xy dy = 0$, given that $y = 1$ when $x = 1$. [PYQ 2018] [Confidence: Very High]

Q37. Solve the linear differential equation: $\frac{dy}{dx} - 3y \cot x = \sin 2x$, given that $y = 2$ when $x = \frac{\pi}{2}$. [PYQ 2017] [Confidence: High]

Q38. Solve the homogeneous differential equation: $x\frac{dy}{dx} = y - x \tan\left(\frac{y}{x}\right)$. [PYQ 2023] [Confidence: Very High]

Q39. Solve the linear differential equation: $(1 + y^2)dx = (\tan^{-1} y - x)dy$. [PYQ 2019] [Confidence: High]

Q40. Solve the linear differential equation: $(x + 1)\frac{dy}{dx} - y = e^{3x}(x + 1)^2$. [PYQ 2020] [Confidence: Very High]

Q41. Solve the homogeneous differential equation: $x^2\frac{dy}{dx} + y(x + y) = 0$, given that $y = 1$ when $x = 1$. [PYQ 2021] [Confidence: High]

Q42. Find the particular solution of the differential equation: $\frac{dy}{dx} + 2y \tan x = \sin x$, given that $y = 0$ when $x = \frac{\pi}{3}$. [Practice] [Confidence: Medium]

5-Mark (Long Answer/Case-Study)

Q43. Find the particular solution of the homogeneous differential equation: $2ye^{x/y}dx + (y - 2xe^{x/y})dy = 0$, given that $x = 0$ when $y = 1$. [PYQ 2019] [Confidence: Very High]

Q44. Find the general solution of the linear differential equation: $\frac{dx}{dy} + \frac{x}{1+y^2} = \frac{\sin^{-1}y}{1+y^2}$. [PYQ 2018] [Confidence: High]

Q45. Find the particular solution of the differential equation: $x\frac{dy}{dx} + y = x \cos x + \sin x$, given that $y = 1$ when $x = \frac{\pi}{2}$. [PYQ 2020] [Confidence: Very High]

Q46. Solve the differential equation: $(x^3 + x^2 + x + 1)\frac{dy}{dx} = 2x^2 + x$. [PYQ 2017] [Confidence: Medium]

Q47. Find the general solution of the homogeneous differential equation: $x \cos\left(\frac{y}{x}\right)\frac{dy}{dx} = y \cos\left(\frac{y}{x}\right) + x$. [PYQ 2022] [Confidence: High]

Q48. Word Problem: The population of a village increases at a rate proportional to the number of its inhabitants present at any time. If the population of the village was 20,000 in 1999 and 25,000 in the year 2004, what will be the population of the village in 2009? [PYQ 2016] [Confidence: High]

Q49. Case Study: A radioactive substance decays at a rate proportional to the amount of substance present. Let $N(t)$ be the amount of the radioactive substance present at time t . (i) Formulate the differential equation representing this situation. (1 Mark) (ii) Solve the differential equation to find $N(t)$ in terms of the initial amount N_0 at $t = 0$. (2 Marks) (iii) If the half-life of the substance is 100 years, find the decay constant k . (2 Marks) [Practice] [Confidence: Very High]

Q50. Case Study: A bank pays interest by continuous compounding, meaning that the rate of increase of the principal is proportional to the principal present at that time. Let P be the principal at time t and $r\%$ be the annual interest rate. (i) Formulate the differential equation for the principal P . (1 Mark) (ii) Find the general solution of this differential equation. (1 Mark) (iii) If the interest rate is 5% per annum, how much will an initial deposit of ₹1000 amount to in 10 years? (Use $e^{0.5} = 1.648$) (2 Marks) [Practice] [Confidence: Very High]

Answer Key

Q1. Order = 2, Degree = Not defined. **Q2.** I.F. = $e^{\int(1/x)dx} = e^{\log x} = x$. **Q3.** Degree = 3. **Q4.**

$\log|y| = \log|x| + C \implies y = Cx$. **Q5.** $\frac{dy}{dx} - \frac{1}{x}y = 2x^2 \implies$ I.F. = $e^{\int(-1/x)dx} = \frac{1}{x}$. **Q6.**

$x^2 + y^2 = r^2 \implies 2x + 2y\frac{dy}{dx} = 0 \implies x + y\frac{dy}{dx} = 0$. Order = 1. **Q7.** Degree = Not defined (due to the $\sin(dy/dx)$

term). **Q8.** I.F. = $e^{\int \sec x dx} = \sec x + \tan x$. **Q9.** Squaring both sides: $\left(\frac{d^2y}{dx^2}\right)^2 = 1 + \left(\frac{dy}{dx}\right)^2$. Order = 2, Degree = 2. **Q10.**

$\frac{dy}{dx} = e^x \cdot e^y \implies e^{-y}dy = e^x dx \implies -e^{-y} = e^x + C \implies e^x + e^{-y} = C$. **Q11.** Order = 3, Degree = 2. **Q12.**

$\frac{dy}{dx} - \frac{x}{1-x^2}y = \frac{1}{1-x^2} \implies$ I.F. = $e^{\int \frac{-x}{1-x^2}dx} = \sqrt{1-x^2}$. **Q13.** Order = 2, Degree = 1. **Q14.** I.F. = $e^{\int Pdy}$. **Q15.**

$y^2 dy = x^2 dx \Rightarrow \frac{y^3}{3} = \frac{x^3}{3} + C \Rightarrow y^3 - x^3 = 3C$. **Q16.** Option A (Both A and R are true, and R is the correct explanation of A). **Q17.** Option D (A is false because degree is not defined, R is true). **Q18.** $\frac{dy}{dx} + \frac{1}{x \log x} y = \frac{2}{x^2} \Rightarrow$ I.F. = $e^{\int \frac{1}{x \log x} dx} = e^{\log(\log x)} = \log x$. **Q19.** Order = 4. **Q20.** Substitution: $x = vy \Rightarrow \frac{dx}{dy} = v + y \frac{dv}{dy}$.

Q21. $\frac{dy}{1+y^2} = \frac{dx}{1+x^2} \Rightarrow \tan^{-1} y = \tan^{-1} x + C$. **Q22.**

$\frac{dy}{dx} = e^{-y}(e^x + x^2) \Rightarrow e^y dy = (e^x + x^2) dx \Rightarrow e^y = e^x + \frac{x^3}{3} + C$. **Q23.**

$\frac{dy}{dx} = (1+x)(1+y) \Rightarrow \frac{dy}{1+y} = (1+x) dx \Rightarrow \log|1+y| = x + \frac{x^2}{2} + C$. **Q24.**

$2^{-y} dy = 2^x dx \Rightarrow \frac{2^{-y}}{\log 2} = \frac{2^x}{\log 2} + C \Rightarrow 2^x + 2^{-y} = C_1$. **Q25.** I.F. = $e^{\int 1 dx} = e^x$. Sol:

$y \cdot e^x = \int e^{-x} \cdot e^x dx = \int 1 dx \Rightarrow ye^x = x + C$. **Q26.** $\frac{dx}{dy} = x + y \Rightarrow \frac{dx}{dy} - x = y$. I.F. = e^{-y} . Sol:

$xe^{-y} = \int ye^{-y} dy \Rightarrow x = -y - 1 + Ce^y$. **Q27.** $\frac{dy}{y^2} = -4x dx \Rightarrow -\frac{1}{y} = -2x^2 + C$. $y(0) = 1 \Rightarrow C = -1$. Sol:

$\frac{1}{y} = 2x^2 + 1 \Rightarrow y = \frac{1}{2x^2 + 1}$. **Q28.** $\frac{dy}{dx} = \frac{y}{x}(\log(y/x) + 1)$. Since it can be written as a function of (y/x) , it is

homogeneous. **Q29.** $\frac{dy}{\sqrt{4-y^2}} = dx \Rightarrow \sin^{-1}(\frac{y}{2}) = x + C$. **Q30.** Divide by $\tan x \tan y$:

$\frac{\sec^2 x}{\tan x} dx + \frac{\sec^2 y}{\tan y} dy = 0 \Rightarrow \log|\tan x| + \log|\tan y| = \log C \Rightarrow \tan x \tan y = C$. **Q31.** Let

$x + y = t \Rightarrow 1 + \frac{dy}{dx} = \frac{dt}{dx} \Rightarrow \frac{dt}{dx} - 1 = \sin t \Rightarrow \frac{dt}{1+\sin t} = dx \Rightarrow \tan t - \sec t = x + C \Rightarrow \tan(x+y) - \sec(x$

$\cdot$ **Q32.** Equation of parabola: $x^2 = 4ay$. Differentiating: $2x = 4a \frac{dy}{dx} \Rightarrow 4a = \frac{2x}{dy/dx}$. Substitute back:

$x^2 = \left(\frac{2x}{dy/dx}\right)y \Rightarrow x \frac{dy}{dx} = 2y$.

Q33. $\frac{dy}{dx} + \frac{2x}{1+x^2} y = \frac{1}{(1+x^2)^2}$. I.F. = $1 + x^2$. Sol: $y(1+x^2) = \int \frac{1}{1+x^2} dx \Rightarrow y(1+x^2) = \tan^{-1} x + C$. **Q34.**

$\frac{dy}{dx} = \frac{y}{x} + \sqrt{1 + (y/x)^2}$. Let

$y = vx \Rightarrow v + x \frac{dv}{dx} = v + \sqrt{1+v^2} \Rightarrow \frac{dv}{\sqrt{1+v^2}} = \frac{dx}{x} \Rightarrow \log|v + \sqrt{1+v^2}| = \log|x| + \log C \Rightarrow y + \sqrt{x^2 + y^2} = C$

$\cdot$ **Q35.** $\frac{dy}{dx} + \frac{2}{x} y = x \log x$. I.F. = x^2 . Sol: $y \cdot x^2 = \int x^3 \log x dx \Rightarrow yx^2 = \frac{x^4}{4} \log x - \frac{x^4}{16} + C$. **Q36.** $\frac{dy}{dx} = \frac{y^2 - x^2}{2xy}$. Let

$y = vx \Rightarrow x \frac{dv}{dx} = \frac{v^2 - 1}{2v} - v = \frac{-1 - v^2}{2v} \Rightarrow \frac{2v}{1+v^2} dv = -\frac{dx}{x} \Rightarrow \log(1+v^2) = -\log x + \log C \Rightarrow (x^2 + y^2)/x^2 = C/$

$\cdot$ $y(1) = 1 \Rightarrow C = 2$. Sol: $x^2 + y^2 = 2x$. **Q37.** I.F. = $e^{\int -3 \cot x dx} = \csc^3 x$. Sol:

$y \csc^3 x = \int \sin 2x \csc^3 x dx = \int 2 \cot x \csc x dx = -2 \csc x + C$. $y(\pi/2) = 2 \Rightarrow 2 = -2 + C \Rightarrow C = 4$. Sol:

$y \csc^3 x = 4 - 2 \csc x$. **Q38.** Let

$y = vx \Rightarrow v + x \frac{dv}{dx} = v - \tan v \Rightarrow \cot v dv = -\frac{dx}{x} \Rightarrow \log|\sin v| = -\log|x| + \log C \Rightarrow x \sin(y/x) = C$.

Q39. $\frac{dx}{dy} + \frac{1}{1+y^2} x = \frac{\tan^{-1} y}{1+y^2}$. I.F. = $e^{\tan^{-1} y}$. Sol: $xe^{\tan^{-1} y} = \int \frac{\tan^{-1} y e^{\tan^{-1} y}}{1+y^2} dy \Rightarrow xe^{\tan^{-1} y} = e^{\tan^{-1} y} (\tan^{-1} y - 1) + C$.

Q40. $\frac{dy}{dx} - \frac{1}{x+1} y = e^{3x}(x+1)$. I.F. = $\frac{1}{x+1}$. Sol: $y \cdot \frac{1}{x+1} = \int e^{3x} dx \Rightarrow \frac{y}{x+1} = \frac{e^{3x}}{3} + C$. **Q41.** $\frac{dy}{dx} = \frac{-xy - y^2}{x^2}$. Let

$y = vx \Rightarrow x \frac{dv}{dx} = -v - v^2 - v = -v^2 - 2v \Rightarrow \frac{dv}{v^2+2v} = -\frac{dx}{x} \Rightarrow \frac{1}{2} \log \left| \frac{v}{v+2} \right| = -\log x + \log C$. Sol with

condition gives $x + 2y = 3x^2 y$. **Q42.** I.F. = $e^{\int 2 \tan x dx} = \sec^2 x$. Sol: $y \sec^2 x = \int \sin x \sec^2 x dx = \sec x + C$.

$y(\pi/3) = 0 \Rightarrow 0 = 2 + C \Rightarrow C = -2$. Sol: $y \sec^2 x = \sec x - 2$.

Q43. $\frac{dx}{dy} = \frac{2xe^{x/y} - y}{2ye^{x/y}}$. Let

$x = vy \Rightarrow v + y \frac{dv}{dy} = \frac{2ve^{v}-1}{2e^v} \Rightarrow y \frac{dv}{dy} = \frac{-1}{2e^v} \Rightarrow 2e^v dv = -\frac{dy}{y} \Rightarrow 2e^{x/y} = -\log|y| + C$.

$y(0) = 1 \Rightarrow C = 2$. Sol: $2e^{x/y} + \log|y| = 2$. **Q44.** I.F. = $e^{\tan^{-1} y}$. Sol: $xe^{\tan^{-1} y} = \int \frac{\sin^{-1} y e^{\tan^{-1} y}}{1+y^2} dy$. Evaluate integral using

substitution and parts. **Q45.** $\frac{dy}{dx} + \frac{1}{x} y = \cos x + \frac{\sin x}{x}$. I.F. = x . Sol: $yx = \int (x \cos x + \sin x) dx = x \sin x + C$.

$y(\pi/2) = 1 \Rightarrow \pi/2 = \pi/2 + C \Rightarrow C = 0$. Sol: $yx = x \sin x \Rightarrow y = \sin x$. **Q46.** $\frac{dy}{dx} = \frac{2x^2 + x}{(x+1)(x^2+1)}$. Use partial

fractions: $\frac{A}{x+1} + \frac{Bx+C}{x^2+1}$. Solve to get $A = 1/2, B = 3/2, C = -1/2$. Integrate:

$y = \frac{1}{2} \log|x+1| + \frac{3}{2} \log(x^2+1) - \frac{1}{2} \tan^{-1} x + C$. **Q47.** Let

$y = vx \Rightarrow v + x \frac{dv}{dx} = \frac{v \cos v + 1}{\cos v} \Rightarrow x \frac{dv}{dx} = \frac{1}{\cos v} \Rightarrow \cos v dv = \frac{dx}{x} \Rightarrow \sin(y/x) = \log|x| + C$. **Q48.**

$\frac{dP}{dt} = kP \Rightarrow P(t) = P_0 e^{kt}$. $P(0) = 20000, P(5) = 25000 \Rightarrow e^{5k} = \frac{5}{4}$. For 2009, $t = 10$.

$P(10) = 20000e^{10k} = 20000(e^{5k})^2 = 20000(25/16) = 31,250$. **Q49.** (i) $\frac{dN}{dt} = -kN$. (ii)

$\frac{dN}{N} = -k dt \Rightarrow N(t) = N_0 e^{-kt}$. (iii) $N(100) = \frac{N_0}{2} \Rightarrow \frac{1}{2} = e^{-100k} \Rightarrow 100k = \log_e 2 \Rightarrow k = \frac{\log_e 2}{100}$. **Q50.** (i)

$\frac{dP}{dt} = \frac{r}{100} P$. (ii) $P(t) = P_0 e^{rt/100}$. (iii) $r = 5 \Rightarrow P(t) = 1000e^{5t/100} = 1000e^{0.05t}$. For $t = 10$,

$P(10) = 1000e^{0.5} = 1000 \times 1.648 = ₹1648$.